Counterexamples to two conjectures of Graffiti on graph eigenvalues

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Abstract

Graffiti, Fajtlowicz's conjecture-making program, produced hundreds of graph-theoretic conjectures, catalogued in *Written on the Wall*. We refute two of its spectral conjectures, both of which survived the 1990–91 computational attack of Brewster, Dinneen and Faber (exhaustive only through ten-vertex graphs) and an eight-algorithm search by Roucairol and Cazenave (2025). Conjecture 143 asserts that in every connected graph the variance of the positive adjacency eigenvalues is at most size/(average distance). It fails on dumbbell graphs (two cliques joined by a path): the smallest counterexample certified here has 39 vertices under both conventions for average distance (37 under the distinct-pairs convention alone), and the ratio of the two sides tends to 2 along a dumbbell sequence. Conjecture 154 asserts, under the standard-deviation reading of "deviation of eigenvalues", that the deviation is at most n/(average distance). It fails on lollipop graphs, at 118 vertices for distinct pairs and 120 for both conventions, with unbounded ratio. All counterexamples carry exact rational certificates, verified in exact arithmetic by two algorithmically independent routes per result (for Conjecture 154, by two independently written checkers).

1 Introduction

Graffiti, written by Siemion Fajtlowicz in the mid-1980s, generates graph-theoretic conjectures on a large scale [5, 4]. Its conjectures — inequalities between graph invariants — were collected, numbered, and annotated by Fajtlowicz in a working document known as *Written on the Wall* [6][1]; the numbering exceeds 800 in the compilation used here, the file archived by Roucairol and Cazenave as `wow-july2004.pdf` (its filename suggests July 2004, but the PDF's own metadata records a creation date of 26 July 2006; we therefore pin the file by its checksum in Section 4 rather than by date). The list attracted substantial attention. Favaron, Mahéo and Saclé [7] resolved many of the early spectral conjectures, proving some and constructing infinite families of counterexamples to others; Brewster, Dinneen and Faber [2] ran a computational attack on roughly 200 of the conjectures and refuted more than 40. The systematic formalization of such invariant inequalities was carried further in the AutoGraphiX line of work of Aouchiche, Hansen and coauthors; see the survey [1]. Most recently, Roucairol and Cazenave applied Monte-Carlo search to spectral graph conjectures, among them Graffiti's Conjecture 137 [9], and then attacked the Graffiti list with a portfolio of eight search algorithms [10]; the results table of [10] lists WoW

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[1] Numbering throughout this note follows *Written on the Wall* (WoW). The later, separate list Graffiti.pc of DeLaViña has its own numbering; its conjecture 143 is unrelated to the one treated here.

Conjecture 143 as open after searching graphs and trees up to size 100, and Conjecture 154 as open after searching up to size 50.

This note refutes Conjecture 143, and refutes Conjecture 154 under the standard-deviation reading of "deviation" fixed in Section 3.

> *"143. variance of positive eigenvalues $\leq$ size / average distance."*
>
> *"154. deviation of eigenvalues $\leq$ n / average distance."*

Here, by the conventions fixed in the WoW preamble ("G always denotes a graph, n the number of its vertices and eigenvalues (unless it is explicitly stated) denote eigenvalues of the adjacency matrix of G"), eigenvalues are those of the adjacency matrix; "size" is the number m of edges; and both statements implicitly concern connected graphs, since the average distance must be defined. Both conjectures predate the 1990–91 computational attack of [2]: the WoW compilation records them among the conjectures that survived it. (That attack was exhaustive over the graphs on at most ten vertices, so survival there carries limited information: the counterexamples exhibited below have at least 37 vertices.) Conjecture 143 fails on *dumbbells* (two cliques joined by a path), with the smallest counterexample certified here on $n = 39$ vertices — and on $n = 37$ vertices under the distinct-pairs convention for average distance (Section 2). Conjecture 154, under the standard-deviation reading of "deviation" discussed in Section 3, fails on *lollipops* (a clique with a pendant path), first within that family at $n = 118$ — at $n = 120$ under both conventions — and along an infinite sequence; the violation there reduces to a pure integer inequality, $8mW^2 > n^5(n-1)^2$, in the Wiener index W. In both cases the failure is structural rather than marginal: along suitable parameter sequences the ratio of left- to right-hand side tends to 2 (Proposition 2.4) and to infinity (Proposition 3.5), respectively. Both refutations are robust to the standard ambiguity in the term "average distance": they hold under the two readings in use in this literature (see below). A literature check (most recently June 2026) found no prior or concurrent resolution of either conjecture; arXiv:2409.18626 [10] remains at version 1, listing both as open.

Conventions. Let G be a connected graph with n vertices, m edges, adjacency eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, and Wiener index $W = \sum_{\{u,v\}} d(u,v)$ (sum over unordered pairs of distinct vertices). Two readings of *average distance* occur in the relevant literature:

$$\ell_{\mathrm{d}} = \frac{2W}{n(n-1)} \qquad \text{and} \qquad \ell_{\mathrm{a}} = \frac{2W}{n^2},$$

the mean over distinct (ordered or unordered) pairs, and the mean over all n^2 entries of the distance matrix, zero diagonal included. The distinct-pairs convention ℓ_{d} is the one used by Favaron, Mahéo and Saclé [7] and matches WoW's own usage (its comment on conjecture 536, asserting that Paley graphs have average distance 3/2, holds only under ℓ_{d}); the all-entries convention ℓ_{a} is the one hard-coded in the public search code of [10]. Since $\ell_{\mathrm{a}} < \ell_{\mathrm{d}}$, right-hand sides computed with ℓ_{a} are *larger*, so the ℓ_{a} reading is the harder one to violate. We refute both readings of both conjectures; only the smallest certified instances depend on the reading (Conjecture 143: $n = 39$ for both readings, $n = 37$ under ℓ_{d} alone; Conjecture 154: $n = 118$ under ℓ_{d}, $n = 120$ for both). "Variance" and "deviation" are the population variance and population standard deviation (divide by the number of data points, not by one less); WoW computes statistical invariants on uniform sample spaces, and the only deviation routine published in the search-code repository of [10] uses the same convention (see Section 3). For Conjecture 154 the reading of "deviation" carries genuine interpretive risk, which we discuss — and quantify — in Section 3.1.

Methods and AI-use disclosure. The counterexamples were found by Demonstrandum, a verification-first multi-agent AI search-and-reasoning pipeline built on Claude language models (Anthropic): agents fixed the conjecture statements and conventions from the frozen source of Section 4, proposed candidate graph families, and ran parameter scans. Certificates were then verified by exact-arithmetic checkers written independently of the discovery code: for Conjecture 143, a single clean-room checker that accepts only if two algorithmically independent exact routes agree; for Conjecture 154, two independently written checkers, the second authored by an adversarial referee agent running on a different model family (Codex, GPT-5.5, OpenAI). All accept paths use integer or rational arithmetic only; floating point appears solely in advisory cross-checks. Both verification bundles passed mutation testing (deliberately corrupted certificates are rejected) and an adversarial automated audit. The text of this note was likewise drafted with the Claude-based pipeline; the author directed the work and takes responsibility for its content. Every number appearing in this note was recomputed during its preparation (June 2026): the certificate checkers and mutation tests were re-run, the family scans were re-executed, and the headline quantities were additionally recomputed by separate scripts (Section 4). The mathematical claims rest on the certificates and checkers described in Section 4 and on the self-contained proofs of Propositions 2.4 and 3.5.

2 Conjecture 143: variance of the positive eigenvalues

For a connected graph G let $k = k(G)$ be the number of strictly positive adjacency eigenvalues, counted with multiplicity, let μ be their mean, and let

$$\mathrm{Var}^+(G) \;=\; \frac{1}{k} \sum_{i:\, \lambda_i > 0} (\lambda_i - \mu)^2$$

be their population variance. Conjecture 143 asserts that $\mathrm{Var}^+(G) \le m/\ell$ for every connected G, with ℓ the average distance.

For integers $t_1, t_2 \ge 1$ and $p \ge 0$, the *dumbbell* $\mathrm{D}(t_1, p, t_2)$ is obtained from disjoint cliques K_{t_1} and K_{t_2} and a path on p internal vertices by joining one vertex of K_{t_1} to one end of the path and the other end of the path to one vertex of K_{t_2} (for $p = 0$, a single bridge edge). It is connected, with $n = t_1 + t_2 + p$ and $m = \binom{t_1}{2} + \binom{t_2}{2} + p + 1$.

Theorem 2.1. *Conjecture 143 is false, under either reading of average distance. The five dumbbells of Table 1 satisfy the indicated strict violations; in particular, on $n = 39$ vertices,*

$$\begin{aligned}\mathrm{Var}^+(\mathrm{D}(7,12,20)) \;=\; 34.31513046745\ldots \;&>\; \frac{m\,n^2}{2W} = \frac{10647}{311} = 34.2347266881\ldots \\ &>\; \frac{m\,n(n-1)}{2W} = \frac{10374}{311},\end{aligned}$$

so both readings fail at $n = 39$; and on $n = 37$ vertices

$$\mathrm{Var}^+(\mathrm{D}(6,12,19)) \;=\; 30.31239262097\ldots \;>\; \frac{m\,n(n-1)}{2W} = \frac{14726}{487} = 30.23819301848\ldots,$$

so the distinct-pairs reading fails at $n = 37$.

Proof. A finite exact computation, certified as described in Section 4. The integers n, m, W, k and the rational right-hand sides $m/\ell_{\mathrm{a}} = mn^2/(2W)$ and $m/\ell_{\mathrm{d}} = mn(n-1)/(2W)$ are elementary exact counts. For Var^+, the certificate provides a rational enclosure of width below 10^{-19}, obtained by exact real-root isolation of the integer characteristic polynomial (no floating point on the accept path); the margin columns of Table 1 are decimal truncations of certified exact rational lower bounds on $\mathrm{Var}^+ - m/\ell$. The clean-room checker accepts the certificate only after two algorithmically independent exact routes certify every claimed inequality (Section 4). $\square$

G	n	m	W	k	$\mathrm{Var}^+(G)$	m/ℓ_{a}	margin	m/ℓ_{d}	margin
D(20, 8, 20)	48	389	6568	6	69.6349632447	56016/821	+1.405974206	54849/821	+2.827411478
D(8, 12, 20)	40	231	5372	8	34.8927644618	46200/1343	+0.492168780	45045/1343	+1.352183672
D(7, 12, 20)	39	224	4976	8	34.3151304675	10647/311	+0.080403779	10374/311	+0.958217284
D(6, 12, 20)	38	218	4581	8	33.9573790158	157396/4581	—	153254/4581	+0.503111388
D(6, 12, 19)	37	199	4383	8	30.3123926210	272431/8766	—	14726/487	+0.074199602

Table 1: Certified dumbbell counterexamples to Conjecture 143. $\mathrm{Var}^+(G)$ is the certified value rounded to ten decimals (the certificate encloses it in a rational interval of width $< 10^{-19}$). The right-hand sides $m/\ell_{\mathrm{a}} = mn^2/(2W)$ and $m/\ell_{\mathrm{d}} = mn(n-1)/(2W)$ are exact. Each margin is a decimal truncation of a certified exact rational lower bound on $\mathrm{Var}^+(G) - m/\ell$; a dash means the corresponding reading is *not* violated by that instance — certified as well: the checker verifies that the certified enclosure of $\mathrm{Var}^+(G)$ lies strictly below that right-hand side.

Remark 2.2 (Minimality within the family; smaller graphs)*.* A floating-point scan of the entire dumbbell family with $n \leq 48$ (all $t_1 \geq 1$, $t_2 \geq t_1$, $p \geq 0$) indicates that $\mathrm{D}(7, 12, 20)$ is the family-minimal violator for the ℓ_{a} reading and $\mathrm{D}(6, 12, 19)$ for the ℓ_{d} reading, with no dumbbell on $n \leq 36$ vertices coming within 0.3 of either bound (the closest approach, by $\mathrm{D}(6, 12, 18)$ on $n = 36$ vertices, falls short by about 0.355) — far above floating-point error, though this scan, unlike Table 1, is not an exact certificate. No minimality over *all* connected graphs is claimed: a simulated-annealing probe at $n = 35$–38 found no smaller violator, but that evidence is heuristic. The prior search of [10] covered sizes up to 100; the counterexamples were found not beyond the searched range but inside it, in a region apparently unfavourable to edge-by-edge search heuristics (a dumbbell must pass through long, low-scoring two-clique-plus-path intermediate states).

Remark 2.3 (Robustness of the reading)*.* Replacing the population variance by the sample variance multiplies the left-hand side by $k/(k-1) > 1$, so all violations in Table 1 persist a fortiori. Reading "size" as n instead of m also leaves the violations intact, since $n < m$ for every instance in the table and the right-hand side only shrinks. The adversarial audit (Section 4) additionally reported, at floating-point level, that replacing average distance by average eccentricity does not rescue the inequality on these instances. The only readings found to rescue it are not supported by the source (e.g. taking the variance of *all* eigenvalues).

The mechanism behind the failure is simple: a dumbbell has exactly two large positive eigenvalues, close to $t - 1$, coming from the cliques, while its remaining positive eigenvalues stay below 3; making the path long drives the variance of the positive spectrum up faster than the ratio m/ℓ. This intuition can be made precise, and shows that the conjecture fails by a factor approaching 2:

Proposition 2.4. *For $t \to \infty$ let $p = p(t)$ be integers with $p \to \infty$ and $p/t \to 0$ (for instance $p = \lfloor t^{1/3} \rfloor$), and let $G_t = \mathrm{D}(t, p, t)$. Then, with $R = m/\ell$ for either reading $\ell \in \{\ell_{\mathrm{a}}, \ell_{\mathrm{d}}\}$,*

$$\frac{\mathrm{Var}^+(G_t)}{R(G_t)} \longrightarrow 2, \qquad \textit{and} \qquad \mathrm{Var}^+(G_t) - R(G_t) \;=\; \big(1 + o(1)\big)\,\frac{2t^2}{p+3} \longrightarrow \infty.$$

In particular G_t violates Conjecture 143, under both readings, for all sufficiently large t.

Proof. All asymptotic notation refers to $t \to \infty$, and all implied constants are absolute; recall $n = 2t + p$ and assume throughout $4 \le p \le t$. Write A for the adjacency matrix of G_t and A_0 for that of the disjoint union $K_t \sqcup P_p \sqcup K_t$, so that $A = A_0 + E$, where E is the adjacency matrix of the two bridge edges. Since $p \ge 2$, the two bridge edges are vertex-disjoint, so E has eigenvalues $1, 1, -1, -1$ and 0 ($n - 4$ times); in particular $\|E\|_2 = 1$.

The spectrum of A_0 is classical [3]: $t - 1$ with multiplicity 2, -1 with multiplicity $2(t - 1)$, and $2\cos\big(j\pi/(p + 1)\big)$ for $j = 1, \dots, p$. Hence for $t \ge 4$ the two largest eigenvalues of A_0 equal $t - 1$, the third largest is $2\cos\big(\pi/(p + 1)\big) < 2$, and the number of positive eigenvalues of A_0 is $k_0 = 2 + \lfloor p/2 \rfloor$.

By Weyl's perturbation inequality, $|\lambda_i(A) - \lambda_i(A_0)| \le \|E\|_2 \le 1$ for all i; therefore

$$\lambda_1(A),\, \lambda_2(A) \in [\,t - 2,\ t\,], \qquad \lambda_i(A) < 3 \quad (i \ge 3). \tag{1}$$

Moreover, since E and $-E$ each have exactly two positive eigenvalues, the Weyl inequalities $\lambda_{i+j-1}(X+Y) \le \lambda_i(X) + \lambda_j(Y)$ (applied with $j = 3$, once to $X = A_0$, $Y = E$ and once to $X = A$, $Y = -E$) give $\lambda_{i+2}(A) \le \lambda_i(A_0)$ and $\lambda_{i+2}(A_0) \le \lambda_i(A)$ for all admissible i. Consequently the number k of (strictly) positive eigenvalues of A satisfies $k \le k_0 + 2$ — if $\lambda_i(A) > 0$ for some $i \ge 3$, then $\lambda_{i-2}(A_0) \ge \lambda_i(A) > 0$, so $i - 2 \le k_0$ — and $k \ge k_0 - 2$, since $\lambda_i(A) \ge \lambda_{i+2}(A_0) > 0$ for every $i \le k_0 - 2$. Hence

$$|k - k_0| \le 2, \qquad \text{so} \qquad k = \tfrac{p}{2} + O(1). \tag{2}$$

Let S_1 and S_2 denote the sum and the sum of squares of the positive eigenvalues of A. By (1) and (2), the $k - 2$ eigenvalues other than λ_1, λ_2 contribute at most $3(k - 2) = O(p)$ to S_1 and at most $9(k - 2) = O(p)$ to S_2, while $\lambda_1 + \lambda_2 = 2t + O(1)$ and $\lambda_1^2 + \lambda_2^2 = 2t^2 + O(t)$. Hence $S_1 = 2t + O(p)$ and $S_2 = 2t^2 + O(t)$, and

$$\mathrm{Var}^+(G_t) = \frac{kS_2 - S_1^2}{k^2} = \frac{2t^2(k-2) + O(tp)}{k^2} = \frac{4t^2}{p}\Big(1 + O\big(\tfrac{1}{p} + \tfrac{p}{t}\big)\Big), \tag{3}$$

using (2) in the last step.

On the other side, $m = t(t - 1) + p + 1 = t^2\big(1 + O(1/t)\big)$. Every distance in G_t is at most the diameter $p + 3$, and the $(t - 1)^2$ pairs formed by one non-attachment vertex from each clique are at distance exactly $p + 3$; splitting the remaining pairs into within-clique pairs (distance 1, at most $t(t - 1)$ in total) and pairs involving a path vertex (fewer than pn of them) gives

$$(t-1)^2(p+3) \;\le\; W \;\le\; t^2(p+3) + t(t-1) + pn(p+3),$$

whence $W = t^2(p+3)\big(1 + O(1/p + p/t)\big)$. Since $n^2 = 4t^2\big(1 + O(p/t)\big)$ and $n(n-1) = n^2\big(1 + O(1/t)\big)$,

$$R(G_t) \;=\; \frac{m\,n^2}{2W} \;\;\text{or}\;\; \frac{m\,n(n-1)}{2W} \;=\; \frac{2t^2}{p+3}\Big(1 + O\big(\tfrac{1}{p} + \tfrac{p}{t}\big)\Big). \tag{4}$$

Dividing (3) by (4):

$$\frac{\mathrm{Var}^+(G_t)}{R(G_t)} = 2 \cdot \frac{p+3}{p}\Big(1 + O\big(\tfrac{1}{p} + \tfrac{p}{t}\big)\Big) = 2 + O\big(\tfrac{1}{p} + \tfrac{p}{t}\big) \longrightarrow 2,$$

since $p \to \infty$ and $p/t \to 0$. Finally $\mathrm{Var}^+ - R = R\,(\mathrm{Var}^+/R - 1) = (1+o(1))\,2t^2/(p+3) \to \infty$, because $t^2/(p+3) \geq t^2/(t+3) \to \infty$. □

As a numerical illustration (floating point, not part of the certificates): D(40, 10, 40) on $n = 90$ vertices has $m = 1571$, $W = 27625$, $k = 7$, $\mathrm{Var}^+ \approx 290.01$ against $m/\ell_{\mathrm{a}} \approx 230.32$ — a violation margin of about 59.7 and a ratio of about 1.26; for D(800, 40, 800) the ratio reaches about 1.81.

3 Conjecture 154: deviation of the eigenvalues

Conjecture 154 reads, verbatim from the decoded WoW source: *"deviation of eigenvalues ≤ n / average distance."* No surviving source defines "deviation" next to the conjecture itself, so the reading must be fixed from context. WoW uses *variance* (e.g. conjectures 38, 143) and *deviation* (e.g. conjectures 39, 40, 140, 154, 244, 253) as distinct invariants and spells out "standard deviation" for degree sequences (conjectures 27, 812, 813); Favaron, Mahéo and Saclé [7], the historical treatment of WoW's eigenvalue-deviation conjectures, define the eigenvalue deviation (there for the Laplacian spectrum) by a root-mean-square formula — a standard deviation, not a mean absolute deviation; and the only public machine formalization, the search code of [10], scores the conjecture by the standard deviation of the adjacency spectrum (its scoring function calls a statistics helper module that is absent from the published tree; the one deviation routine the repository does publish, in `tools/calc.rs`, is a population standard deviation). We therefore work under the **standard-deviation reading**, and claim the refutation *only* under that reading; Section 3.1 states exactly what was checked about the alternative (mean-absolute-deviation) reading.

Under the standard-deviation reading the left-hand side is exactly computable from n and m alone:

Lemma 3.1. *For every graph G with n vertices and m edges, the population standard deviation of the n adjacency eigenvalues equals* $\mathrm{dev}(G) = \sqrt{2m/n}$.

Proof. $\sum_i \lambda_i = \operatorname{tr} A = 0$ and $\sum_i \lambda_i^2 = \operatorname{tr} A^2 = \sum_v \deg(v) = 2m$, so the population variance is $\frac{1}{n}\sum_i \lambda_i^2 - \big(\frac{1}{n}\sum_i \lambda_i\big)^2 = 2m/n$. □

Lemma 3.2. *Let G be connected with $n \geq 2$ vertices, m edges and Wiener index W. Under the standard-deviation reading, G violates Conjecture 154*

- *under the distinct-pairs convention $\ell = \ell_{\mathrm{d}}$ if and only if $8mW^2 > n^5(n-1)^2$;*
- *under the all-entries convention $\ell = \ell_{\mathrm{a}}$ if and only if $8mW^2 > n^7$.*

Proof. By Lemma 3.1 the conjecture asserts $\sqrt{2m/n} \leq n/\ell$. Both sides are positive, so the inequality is equivalent to its square, $2m/n \leq n^2/\ell^2$. Substituting $\ell_{\mathrm{d}} = 2W/(n(n-1))$ gives $2m/n \leq n^4(n-1)^2/(4W^2)$, i.e. $8mW^2 \leq n^5(n-1)^2$; substituting $\ell_{\mathrm{a}} = 2W/n^2$ gives $8mW^2 \leq n^7$. Violation is the strict reverse inequality. Note $n^7 > n^5(n-1)^2$, so every violation of the ℓ_{a} reading is automatically one of the ℓ_{d} reading. □

The verdict is thus a pure integer comparison — no eigenvalue computation and no floating point are needed at all. For integers $t \geq 3$, $p \geq 1$, the *lollipop* $\mathrm{L}(t,p)$ is the clique K_t with a pendant path on p vertices attached by a bridge edge to a clique vertex v_0; it is connected with $n = t + p$ and $m = \binom{t}{2} + p$.

Lemma 3.3. *The Wiener index of the lollipop is*

$$W(\mathrm{L}(t,p)) = \binom{t}{2} + \sum_{k=1}^{p} [k + (t-1)(k+1)] + \binom{p+1}{3} = \binom{t}{2} + \frac{p(p+1)}{2} + (t-1)\,\frac{p(p+3)}{2} + \binom{p+1}{3}.$$

Proof. Label the path vertices $u_1, \ldots, u_p$, with u_k at distance k from the attachment vertex v_0. The $\binom{t}{2}$ clique pairs are at distance 1. The pair $\{u_k, v_0\}$ is at distance k and the pairs $\{u_k, w\}$, for the $t-1$ clique vertices $w \neq v_0$, are at distance $k+1$. The path-path pairs contribute $\sum_{1 \leq i < j \leq p}(j-i) = \binom{p+1}{3}$. Summing gives the first expression; evaluating $\sum_{k=1}^{p} k = p(p+1)/2$ and $\sum_{k=1}^{p}(k+1) = p(p+3)/2$ gives the second. $\square$

Theorem 3.4. *Conjecture 154 is false under the standard-deviation reading of "deviation", under both conventions for average distance. The integer data of the three lollipops in Table 2 are exact; by Lemma 3.2,* $\mathrm{L}(48,70)$ *on* $n = 118$ *vertices violates the distinct-pairs reading, and* $\mathrm{L}(50,70)$ *on* $n = 120$ *and* $\mathrm{L}(72,72)$ *on* $n = 144$ *vertices violate both readings.*

Proof. m and W are evaluated by Lemma 3.3 (and were re-verified by breadth-first search over the explicit edge lists); the products and comparisons in Table 2 are exact integer arithmetic, reproduced by two independent implementations (Section 4). $\square$

	$\mathrm{L}(48,70)$	$\mathrm{L}(50,70)$	$\mathrm{L}(72,72)$
n	118	120	144
m	1198	1295	2628
W	180 853	186 060	259 080
$8mW^2$	313 471 628 124 656	358 645 832 496 000	1 411 182 313 113 600
$n^5(n-1)^2$	313 171 159 328 352	352 370 995 200 000	1 266 148 181 016 576
n^7	318 547 390 056 832	358 318 080 000 000	1 283 918 464 548 864
$8mW^2 - n^5(n-1)^2$	+300 468 796 304	+6 274 837 296 000	+145 034 132 097 024
$8mW^2 - n^7$	−5 075 761 932 176	+327 752 496 000	+127 263 848 564 736
violates	ℓ_d only	both	both

Table 2: Integer certificates for Theorem 3.4 (Lemma 3.2: violation of the ℓ_d reading iff $8mW^2 > n^5(n-1)^2$, of the ℓ_a reading iff $8mW^2 > n^7$).

For orientation: $\mathrm{dev}(\mathrm{L}(72,72)) = \sqrt{73/2} = 6.041522\ldots$ against $n/\ell_\mathrm{a} = n^3/(2W) = 62208/10795 = 5.762667\ldots$ and $n/\ell_\mathrm{d} = n^2(n-1)/(2W) = 61776/10795 = 5.722649\ldots$; in the squared comparison of Lemma 3.2 the threshold is exceeded by about 11% (ℓ_d) and 10% (ℓ_a). The whole refutation can be re-verified in well under a minute; the following self-contained check (quoted verbatim from the certificate bundle) needs nothing beyond a Python interpreter:

```
t, p = 72, 72
n, m = t+p, t*(t-1)//2 + p
W = t*(t-1)//2 + sum(k + (t-1)*(k+1) for k in range(1, p+1)) \
    + (p+1)*p*(p-1)//6
```

```
assert 8*m*W*W > n**5 * (n-1)**2     # distinct-pairs convention
assert 8*m*W*W > n**7                # all-n^2-entries convention
print("Graffiti 154 (std-dev reading) refuted:", n, m, W)
```

The failure is again structural: the standard deviation $\sqrt{2m/n}$ is unbounded along lollipops with $p = t$, while n/ℓ stays bounded.

Proposition 3.5. *For every integer $t \geq 71$, the lollipop $\mathrm{L}(t,t)$ on $n = 2t$ vertices violates Conjecture 154 (standard-deviation reading) under both conventions for average distance, and*

$$\frac{\operatorname{dev}(\mathrm{L}(t,t))}{n/\ell} \geq \frac{\sqrt{(t+1)/2}}{6} \longrightarrow \infty \qquad (\ell \in \{\ell_{\mathrm{a}}, \ell_{\mathrm{d}}\}).$$

Proof. Here $m = \binom{t}{2} + t = t(t+1)/2$, so $\operatorname{dev} = \sqrt{2m/n} = \sqrt{(t+1)/2}$ exactly, by Lemma 3.1. Specializing Lemma 3.3 to $p = t$:

$$\begin{aligned} W &= \frac{t(t-1)}{2} + \frac{t(t+1)}{2} + \frac{t(t-1)(t+3)}{2} + \frac{(t+1)t(t-1)}{6} \\ &= \frac{t\,(6t + 3(t-1)(t+3) + (t^2-1))}{6} = \frac{t\,(2t^2+6t-5)}{3}. \end{aligned}$$

Hence

$$\frac{n}{\ell_{\mathrm{d}}} = \frac{n^2(n-1)}{2W} = \frac{6t(2t-1)}{2t^2+6t-5} \qquad \text{and} \qquad \frac{n}{\ell_{\mathrm{a}}} = \frac{n^3}{2W} = \frac{12t^2}{2t^2+6t-5}.$$

Both are smaller than 6 for every $t \geq 1$: indeed $6t(2t-1) < 6(2t^2+6t-5)$ amounts to $30 < 42t$, and $12t^2 < 6(2t^2+6t-5)$ amounts to $30 < 36t$. For $t \geq 71$ we have $\operatorname{dev} = \sqrt{(t+1)/2} \geq 6 > n/\ell$ for both conventions, which is the asserted (strict) violation, and the ratio bound follows. □

Remark 3.6 (Minimality within the family). An exhaustive integer-arithmetic scan of all lollipops $\mathrm{L}(t,p)$ with $t \geq 3$, $p \geq 1$ and $n = t + p \leq 130$ (re-run, like all computations quoted here, while preparing this note; the independent audit re-ran it allowing the degenerate values $t = 1, 2$ as well) shows: no lollipop with $n \leq 117$ violates either reading; at $n = 118$ there are exactly four violators of the ℓ_{d} reading, $\mathrm{L}(48,70)$, $\mathrm{L}(49,69)$, $\mathrm{L}(50,68)$, $\mathrm{L}(51,67)$, none violating the ℓ_{a} reading; at $n = 119$ exactly seven lollipops, $\mathrm{L}(t, 119-t)$ for $47 \leq t \leq 53$, violate the ℓ_{d} reading and again none the ℓ_{a} reading; and at $n = 120$ exactly two lollipops, $\mathrm{L}(50,70)$ and $\mathrm{L}(51,69)$, violate both readings. Since this scan is exact, these statements are certificates, not numerics — but they concern the lollipop family only. The minimum order of a counterexample over all connected graphs is not determined here: a smaller, non-lollipop violator of any order up to 117 is not excluded — the prior search for this conjecture [10] reached size 50, but it was a heuristic search, not an exhaustive enumeration.

3.1 The mean-absolute-deviation reading

If Graffiti's "deviation" meant the *mean absolute deviation* (MAD) rather than the standard deviation, the instances above would prove nothing: we record here, for completeness, exactly what was checked. Since the adjacency spectrum has mean zero, the MAD equals $\frac{1}{n}\sum_i |\lambda_i| = \mathcal{E}(G)/n$, where $\mathcal{E}(G)$ is the graph energy. For each $G \in \{\mathrm{L}(48,70), \mathrm{L}(50,70), \mathrm{L}(72,72)\}$ it was *proved in exact arithmetic* that

$$\frac{\mathcal{E}(G)}{n} < \frac{n}{\ell} \qquad \text{for both } \ell \in \{\ell_{\mathrm{a}}, \ell_{\mathrm{d}}\},$$

i.e. these lollipops do *not* violate the MAD reading. Two independent exact routes were used: (i) a Sturm-certified rational enclosure of $\mathcal{E}(G)$ of width below 3×10^{-17}, computed from the integer characteristic polynomial of the $(p+2) \times (p+2)$ quotient matrix of the equitable partition $\{v_0\}, V(K_t) \setminus \{v_0\}, \{u_1\}, \ldots, \{u_p\}$, together with the eigenvalue -1 of multiplicity $t-2$ (both verified by explicit eigenvectors and the two trace identities); and (ii) an exact-rational application of the Koolen–Moulton energy bound [8]. Numerically, the certified energies are $\mathcal{E} = 183.02566963\ldots$, $187.02616866\ldots$, $233.57628649\ldots$ respectively, giving MAD values $1.5510\ldots$, $1.5585\ldots$, $1.6220\ldots$ against right-hand sides between $4.5039\ldots$ and $5.7626\ldots$ — sign gaps of $+2.95$ to $+4.14$, certified. The refutation of Conjecture 154 in this note is therefore claimed *only* under the standard-deviation reading, which is the reading of [7] and of the only known machine formalization [10]; the residual interpretive risk is thereby documented, quantified, and confined. (For the sample standard deviation, which exceeds the population standard deviation by the factor $\sqrt{n/(n-1)} > 1$, the violations hold a fortiori.)

4 Verification

Both refutations are packaged as machine-checkable certificate bundles, with no floating point on any accept path. Each result is certified by two algorithmically independent exact-arithmetic routes: for Conjecture 143 the two routes live inside a single clean-room checker, while Conjecture 154 has two independently written checkers.

Statement provenance. The conjecture statements were frozen from the compilation of *Written on the Wall* archived as `wow-july2004.pdf` [6] — the same file used by [10], pinned here by the checksum below. That PDF uses Type-3 bitmap fonts without ToUnicode maps; the statements were recovered by a glyph-decoding script validated against the document title and the surrounding conjectures; the statement of Conjecture 154 was additionally confirmed by pixel-level rendering of the relevant pages during the adversarial audit.

Conjecture 143. The certificate lists, for each of the five dumbbells of Table 1, the explicit edge list, the integers n, m, W, k, the exact rational right-hand sides, a rational enclosure of Var^+ of width below 10^{-19}, and exact rational lower bounds for every claimed margin. The checker (written clean-room, independently of the discovery pipeline) verifies every instance by two algorithmically independent exact routes and accepts only if both certify each claimed strict inequality and, for each reading not claimed (the dashes of Table 1), that the certified enclosure of Var^+ lies strictly below the right-hand side: route A computes the integer characteristic polynomial symbolically and isolates its real roots exactly (isolation width 2^{-70}); route B is restricted to the Python standard library (exact fractions only) and recomputes the characteristic polynomial by the Berkowitz algorithm, checks it against an equitable-partition quotient via the exact integer identity $\mathrm{char}_A(x) = \mathrm{char}_B(x)\,(x+1)^{t_1+t_2-4}$, isolates roots by Sturm chains with bisection refinement to width 2^{-70}, and evaluates the variance in rational interval arithmetic. Cross-checks include identical characteristic polynomials, trace and edge-count coefficient identities, agreement of k, intersection of the two variance enclosures, connectivity, and recomputation of W, m and the right-hand sides from the edge lists. Ten targeted certificate corruptions (mutation tests) are all rejected, the pristine certificate accepted. An independent audit recomputed all five instances with separate numerical software and confirmed every certificate field, and an orchestrator-level script rebuilt $\mathrm{D}(20, 8, 20)$ from the construction description alone (rather than from the certified edge list) and reproduced both violations at floating-point precision.

Conjecture 154. Here the accept path is integer-only (Lemma 3.2). Checker A rebuilds the three lollipops from scratch, recomputes W both by breadth-first search and by the closed form of Lemma 3.3 (demanding agreement), evaluates the integer violation forms, re-runs the exhaustive $n \leq 130$ family scan of Remark 3.6, and proves the MAD non-violation of Section 3.1 in exact arithmetic by the two routes described there; the spectral side-checks (trace identities at double and 60-digit precision, agreement of the certified spectrum with a numerical eigensolver) are advisory only. Checker B was written independently by an adversarial referee agent from a different model family (Codex, GPT-5.5) and reproduces the integer inequalities, the margins, and the $n \leq 117$ non-violation from its own reconstruction. Six mutation tests (e.g. claiming a violation at $\mathrm{L}(47, 70)$, corrupting W by one, claiming the ℓ_{a} reading at $\mathrm{L}(48, 70)$) are all rejected. An orchestrator-level script again rebuilds $\mathrm{L}(72, 72)$ from the construction description and re-asserts both integer inequalities.

Independent recomputation for this note. In June 2026, while preparing this note, both checkers and the mutation tests were re-run (verdicts: `ACCEPT` and `ALL CHECKS PASS`, with all mutation tests rejected); these re-runs certify every entry of Tables 1 and 2, the lollipop family scan of Remark 3.6, and the mean-absolute-deviation bounds of Section 3.1. The dumbbell family scan of Remark 2.2 was re-executed, and the headline quantities — the values of Table 1 and the numerical illustrations $\mathrm{D}(40, 10, 40)$, $\mathrm{D}(800, 40, 800)$ and $\mathrm{L}(t, t)$ — were additionally recomputed by separate scripts written for the purpose.

Data availability. The two certificate bundles — machine-readable certificates (JSON), all checkers, the mutation-test suites, the frozen source PDF of *Written on the Wall* with its decoding script, the archived search code of [10] used to fix the formalization conventions, and full verification logs — are available at `https://doi.org/10.5281/zenodo.20673865` and `https://github.com/demonstrandum-research/artifacts`. The frozen source PDF has SHA-256 checksum

`d2c779d2c28418b30ab1b4f84bf7112c0e000bca1f2d6a3c48d0baba78af7733`.

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